Bike Sharing Demand Regression Task

Presented by:
Alex Hines and
Md Zahidul Islam



THE UNIVERSITY OF
TENNESSEE
KNOXVILLE

Presentation Overview

- Review of Project
- Goals from Presentation #2
- Feature Selection
- LSTM and Results
- Hyperparameter Tuning Methodology
- Hyperparameter Tuning Results
- Model Comparison Table
- Future Research

Review of Project

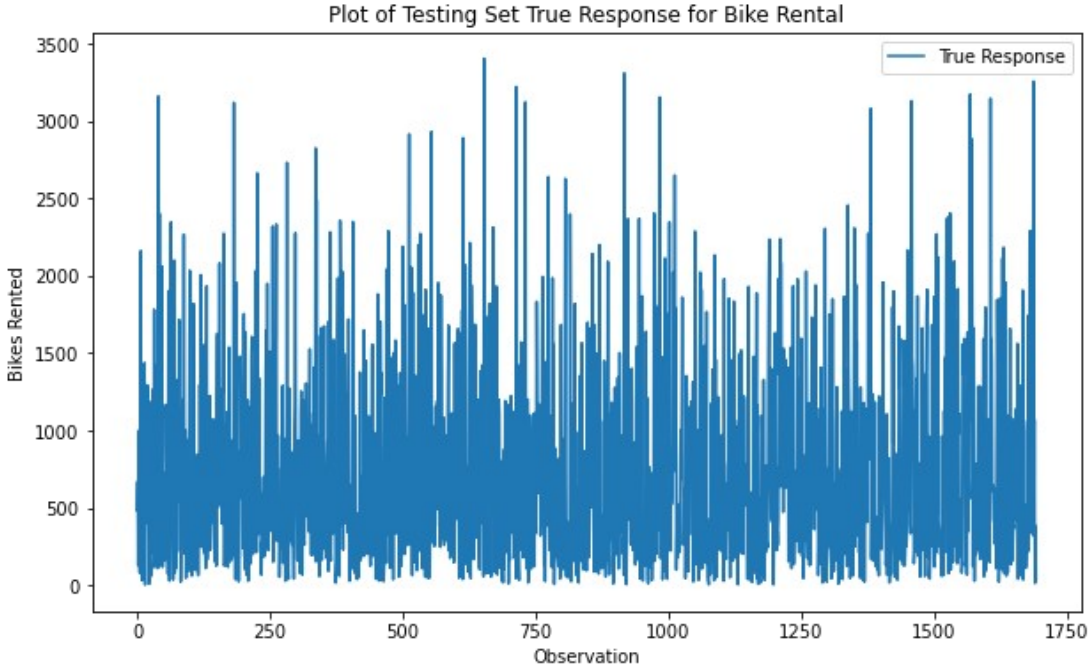
- As we can recall from our last presentation, the data set that we are researching contains two years of data from a bike rental company in Seoul, South Korea. Data is collected 24 hours per day for two years.
- The data set head is shown below and displays our predictor variables as well as our target variable (bike demand).

	Date	Rent_Bike_Count	Hour	Temperature	Humidity	Wind_speed	Visibility	DPT	Solar_Radiation	Rainfall	Snowfall	Seasons	Holiday	Functioning_
0	01/12/2017	254	0	-5.2	37	2.2	2000	-17.6	0.0	0.0	0.0	Winter	No Holiday	
1	01/12/2017	204	1	-5.5	38	0.8	2000	-17.6	0.0	0.0	0.0	Winter	No Holiday	
2	01/12/2017	173	2	-6.0	39	1.0	2000	-17.7	0.0	0.0	0.0	Winter	No Holiday	
3	01/12/2017	107	3	-6.2	40	0.9	2000	-17.6	0.0	0.0	0.0	Winter	No Holiday	
4	01/12/2017	78	4	-6.0	36	2.3	2000	-18.6	0.0	0.0	0.0	Winter	No Holiday	

Review of Project (cont.d)

- Total Bikes Rented: The number of bikes rented during the one-hour period.
- Hour: Hour of operation
- Temperature: Temperature in Celsius
- Humidity: Humidity value in %
- Wind Speed: Wind speed in m/s
- Visibility: Visibility in m
- Dew Point Temperature: Dew Point Temperature in Celsius
- Solar Radiation: Solar Radiation in MJ/m2
- Rainfall: Rainfall in mm
- Snowfall: Snowfall in cm
- Seasons: Winter, Spring, Summer, Autumn
- Holiday: Yes or No
- Functional Day: Yes or No

Bike Demand for Testing Set



Goals from Presentation #2

- Complete work on the LSTM model.
- Complete hyper-parameter tuning on the following models to determine if they are optimal.
  - ANN, XGBoost, Random Forest
- Compare error metrics between models to determine which models, if any, are optimal.

Feature Selection

	Hour	Temperature	Humidity	Wind Speed	Is Visible	DPT	Is Sunny	Is Raining	Is Snowing	Season	Is Holiday	Functioning Day	Bikes Rented
Hour	1.000000	0.124114	-0.241644	0.285197	0.057888	0.003054	0.159470	-0.027163	-0.032174	0.000000	0.000000	0.005439	0.410257
Temperature	0.124114	1.000000	0.159371	-0.036252	0.005500	0.912798	0.278736	0.074539	-0.333068	0.591545	-0.055931	-0.050170	0.538558
Humidity	-0.241644	0.159371	1.000000	-0.336683	-0.222383	0.536894	-0.455671	0.397966	0.051090	0.189238	-0.050278	-0.020800	-0.199780
Wind Speed	0.285197	-0.036252	-0.336683	1.000000	0.037164	-0.176486	0.334140	-0.042979	0.051113	-0.166834	0.023017	0.005037	0.121108
Is Visible	0.057888	0.005500	-0.222383	0.037164	1.000000	-0.077949	-0.100376	-0.110214	-0.026887	0.115041	0.065558	-0.028117	0.103324
DPT	0.003054	0.912798	0.536894	-0.176486	-0.077949	1.000000	0.039187	0.205161	-0.267431	0.582418	-0.066759	-0.052837	0.379788
Is Sunny	0.159470	0.278736	-0.455671	0.334140	-0.100376	0.039187	1.000000	-0.144819	-0.056385	0.066590	-0.003706	-0.009594	0.268261
Is Raining	-0.027163	0.074539	0.397966	-0.042979	-0.110214	0.205161	-0.144819	1.000000	0.002843	0.037805	-0.011161	0.015371	-0.212493
Is Snowing	-0.032174	-0.333068	0.051090	0.051113	-0.026887	-0.267431	-0.056385	0.002843	1.000000	-0.239985	0.007587	0.043084	-0.185897
Season	0.000000	0.591545	0.189238	-0.166834	0.115041	0.582418	0.066590	0.037805	-0.239985	1.000000	-0.057614	-0.196082	0.359687
Is Holiday	0.000000	-0.055931	-0.050278	0.023017	0.065558	-0.066759	-0.003706	-0.011161	0.007587	-0.057614	1.000000	-0.027624	-0.072338
Functioning Day	0.005439	-0.050170	-0.020800	0.005037	-0.028117	-0.052837	-0.009594	0.015371	0.043084	-0.196082	-0.027624	1.000000	0.203943
Bikes Rented	0.410257	0.538558	-0.199780	0.121108	0.103324	0.379788	0.268261	-0.212493	-0.185897	0.359687	-0.072338	0.203943	1.000000

Feature Selection

- After the correlation matrix has been calculated, we will determine which features should be kept in the model.
- First, we will look at the ‘Bikes Rented’ column as this is the target variable in this research. It is important to keep features with high absolute values of correlation values for this specific column, as it will allow us to determine which variables have an impact on the target response.
- The highest correlations for this column are the correlations between ‘Bikes Rented’ and ‘Hour of Day’, ‘Season’, ‘Is Sunny’, and ‘Dew Point Temperature’. These variables were all highly correlated with the bikes rented column, so they are likely good predictors for the model.
- On the other hand, functioning day has little to no correlation with any other variables. This column was removed in Presentation #2, and this supports the idea of removing this feature from the model.

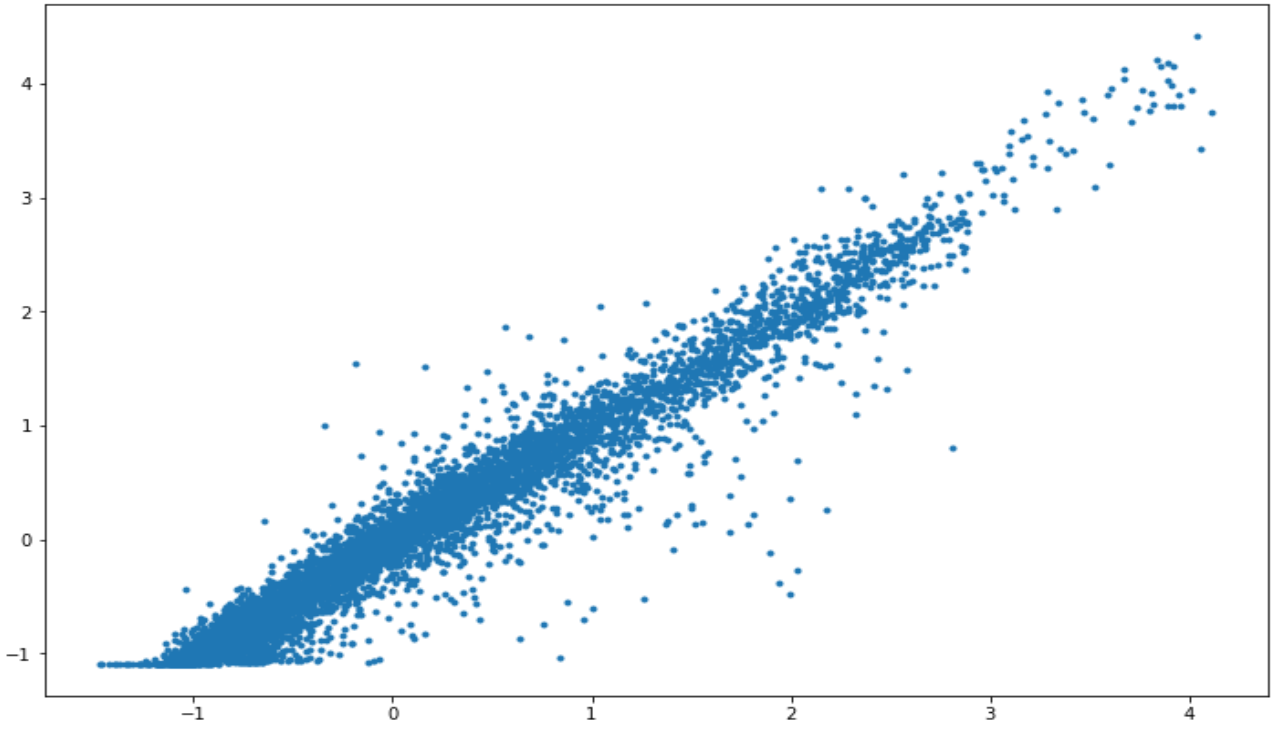
LSTM Model Implementation

- As mentioned in Presentation #2, one of the main goals for Presentation #3 was to develop a Long Short-Term Memory (LSTM) neural network.
- Since this is a time-series regression problem, an LSTM model seems to be a feasible method of creating accurate, reliable predictions for bike rentals.
- We chose to use the PyTorch implementation of the LSTM.
- We divided the dataset into training batches and testing batches using a sliding window. Both training and testing batch sizes were (96, 730, 12). There were 68 training batches and 75 testing batches.

LSTM Model Structure

- Due to the large amount of data in our set, we chose to stick with a basic LSTM network structure shown below. This allows the computation time to not grow even larger.
- The first layer is an input layer with an input size of 12, hidden size of 100, and output size of 1.
- The second layer is a dropout layer of .2.
- Finally, the last layer is a fully connected layer of 100 input features and a single output feature (our prediction).

LSTM Model Results



R-Squared = **0.9520**
RMSE = **151.2414**
MAE = 87.9315

Hyper-Parameter Tuning: ANN

- We used Ray_Tune package for tuning the hyperparameters.
- We varied the size of the layers, learning rate and dropout

The configuration is as follows:

"h1": 10*randint(1,4)

"h2": 10*randint(1,4)

"h3": 10*randint(1,4)

"h4": 10*randint(1,4)

"h5": 10*randint(1,4)

"p": uniform(0,.4)

"lr": loguniform(1e-4, 1e-1)

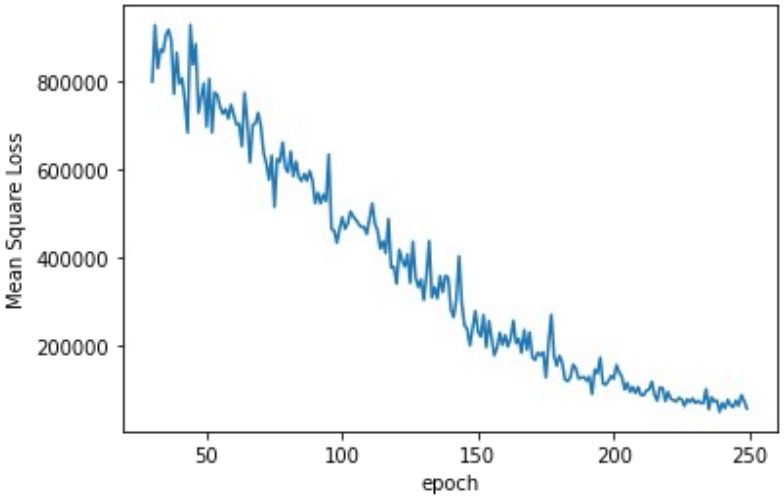
Hyper-Parameter Tuning: ANN

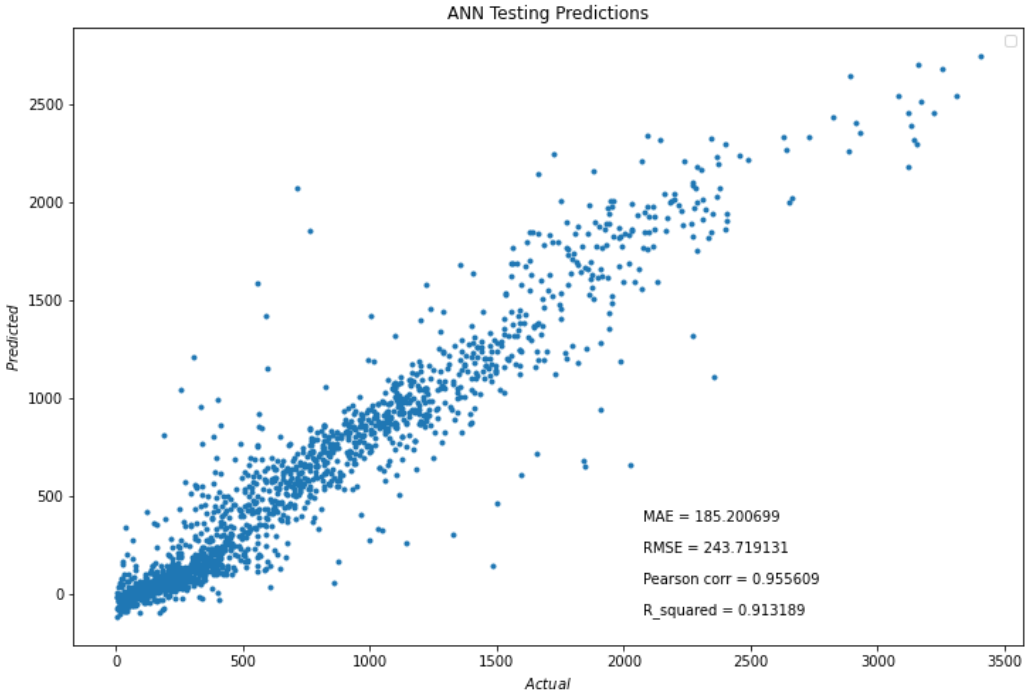
The best trial configuration after 128 trial runs-

'h1': 100,
'h2': 1000,
'h3': 100,
'h4': 100,
'h5': 1000,
'p': 0.017647347432403083
'lr': 0.0386999044944899

We trained the ANN model again with the optimized parameters.

Artificial Neuron Network (ANN)- Before tuning



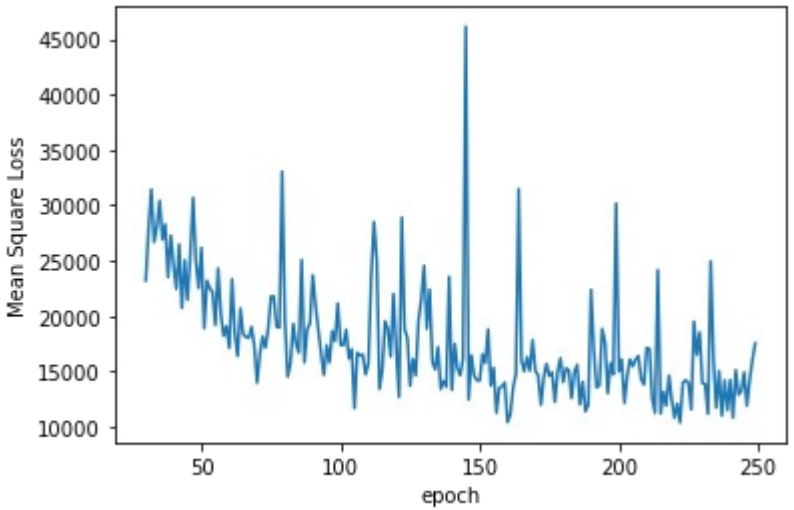


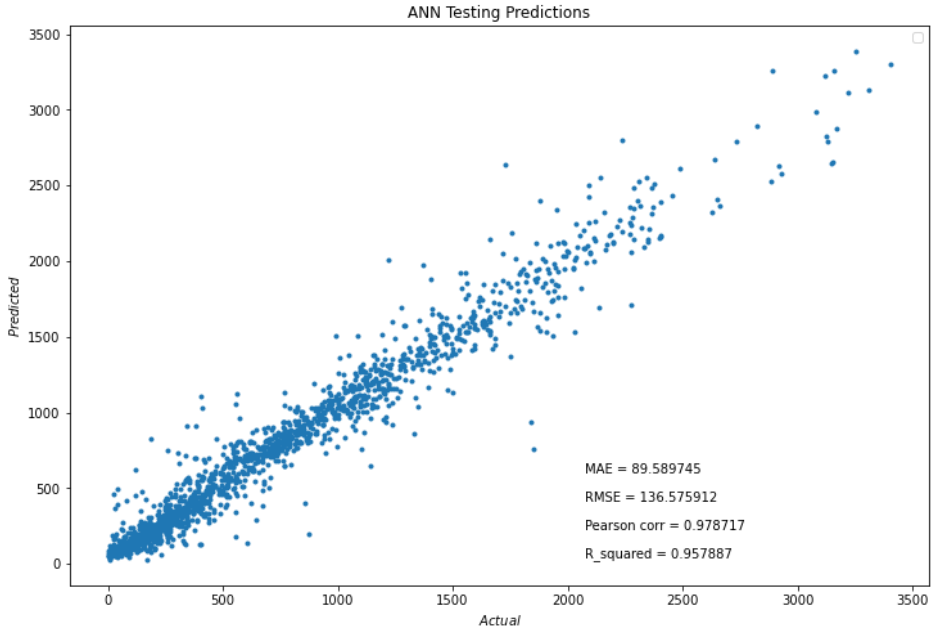
Training Loss: 795537.12
Testing Loss: 799069.50

R-Squared = **0.9131**
RMSE = 243.71
MAE = 185.20

***Good Prediction. Can be improved.*

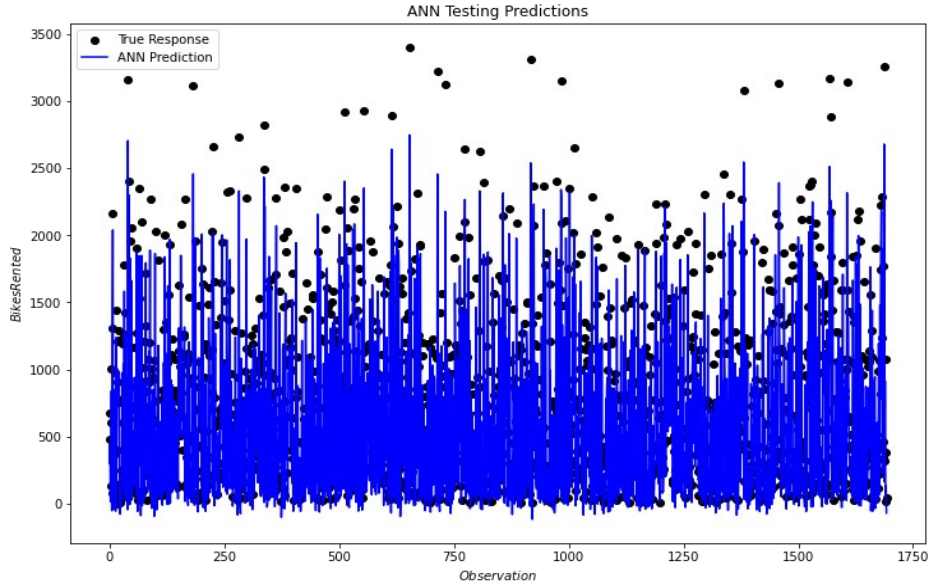
Artificial Neuron Network (ANN)- After tuning



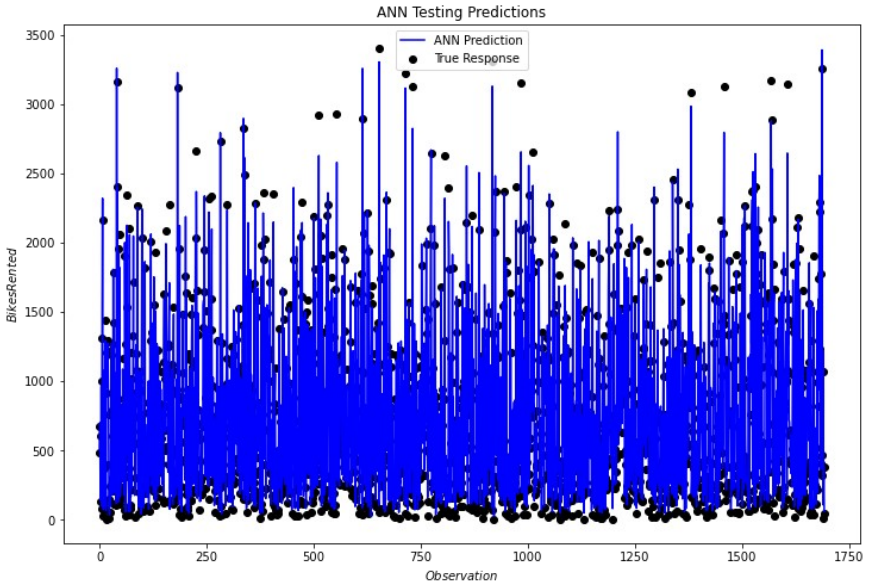


R-Squared = **0.96**
RMSE = 136.57
MAE = 89.58

Artificial Neuron Network (ANN)

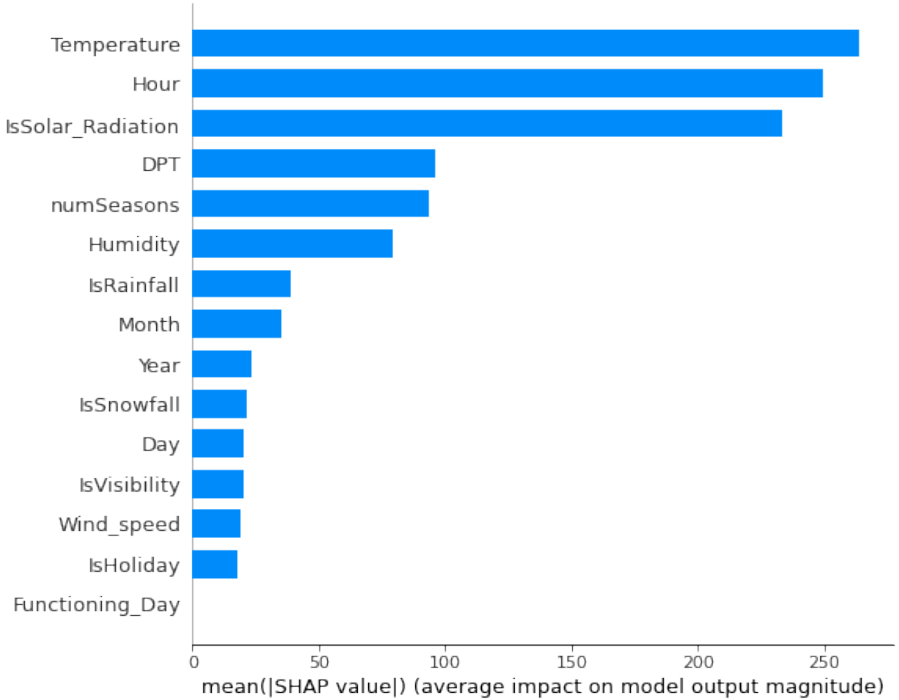


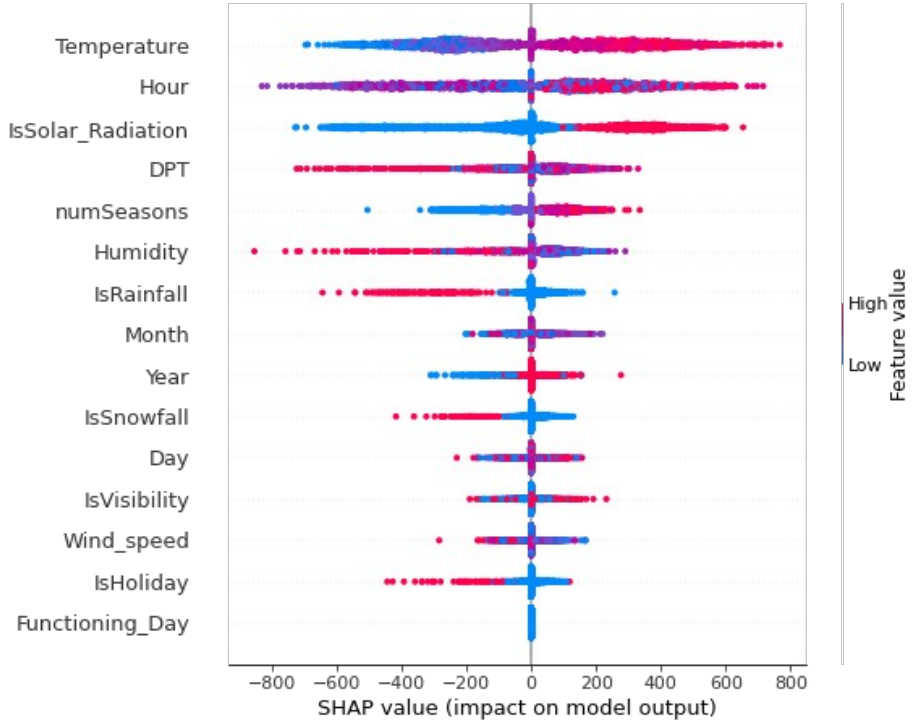
Before Tuning



After Tuning

ANN (Feature Importance)





Hyper-Parameter Tuning: XGBoost

- We used Sequential Grid Search for tuning
- ❑ First, maximum depth was tuned using, max_depths = range(5,15)
- ❑ Then, subsamples, colsample_bytrees, colsample_bylevel was tuned using subsamples = linspace(0.1, 1.0, 10), colsample_bytrees = linspace(0.1, 0.5, 5), colsample_bylevel = linspace(0.1, 0.5, 5).
- ❑ Finally, learing rate was tuned using, learning_rates = logspace(-3, -1, 5).

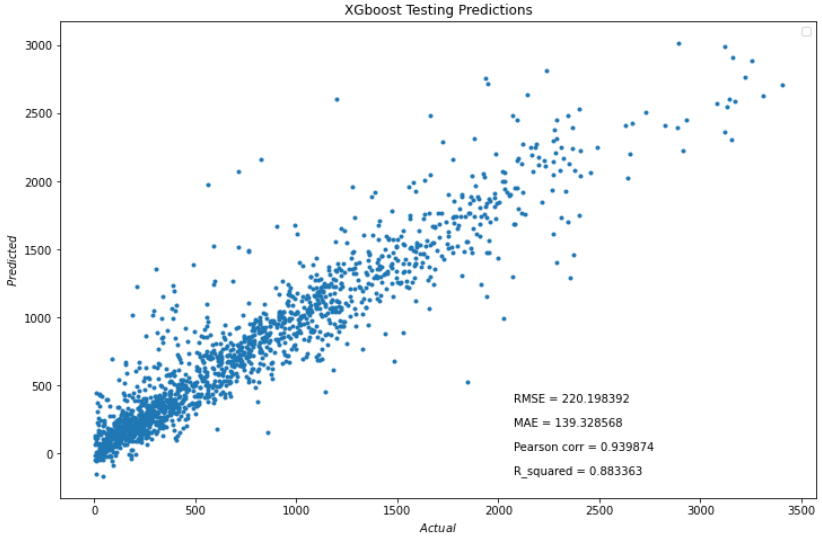
Hyper-Parameter Tuning: XGBoost

The best hyperparameters after tuning were-

colsample_bylevel=0.5,
colsample_bynode=None
colsample_bytree=0.5,
importance_type='gain’
learning_rate=0.01,
max_depth=7,
n_estimators=3438
random_state=42
subsample=0.9

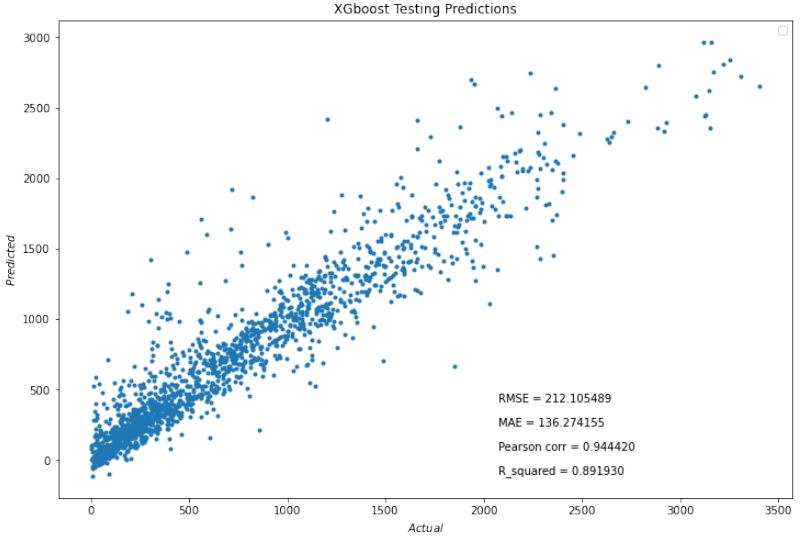
XGBoost

Before Tuning



R-Squared = 0.8833
RMSE = 139.32
MAE = 220.19

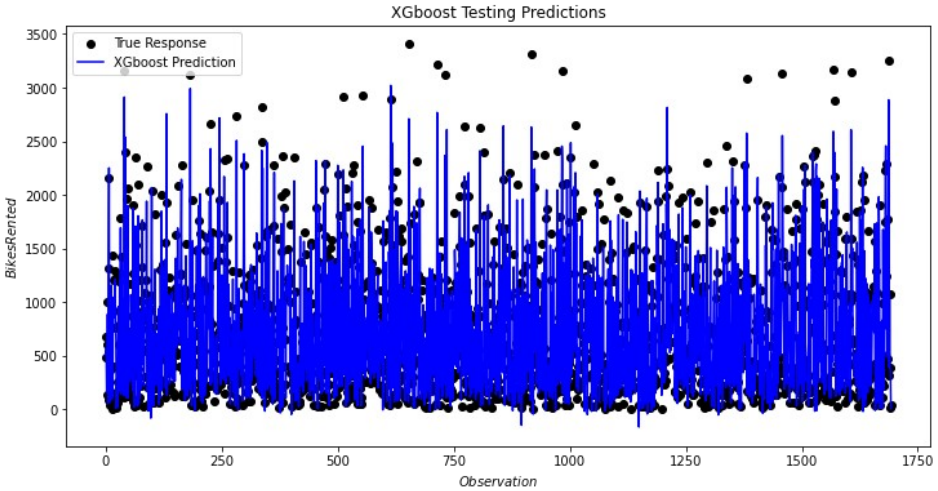
After Tuning



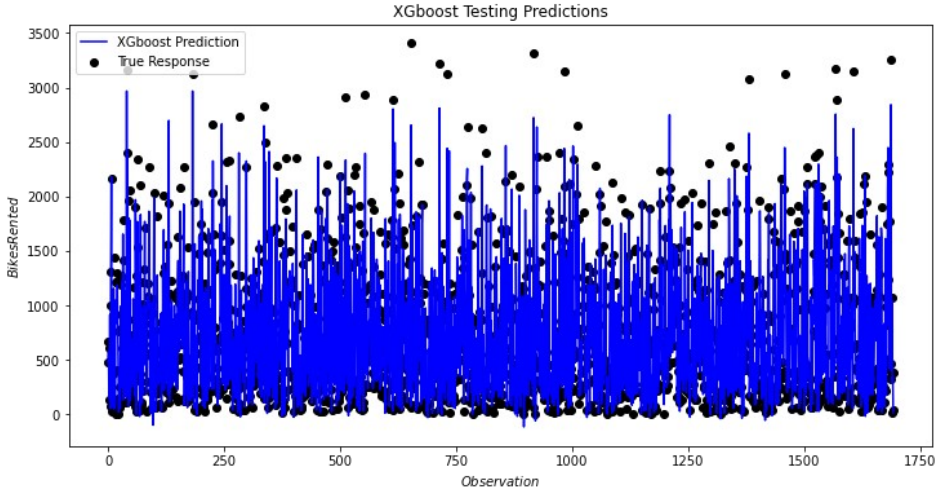
R-Squared = 0.89
RMSE = 136.27
MAE = 212.105

XGBoost

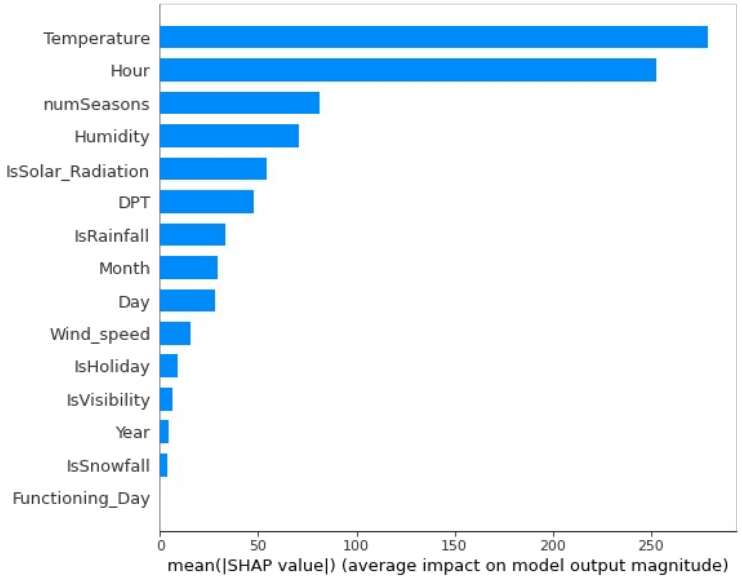
Before Tuning

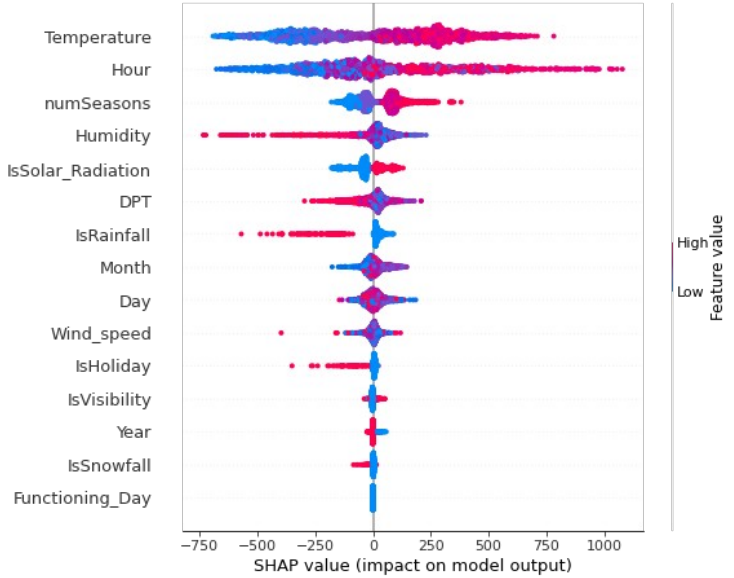


After Tuning



XGBoost (feature Importance)





Hyper-Parameter Tuning: Random Forest

- We used GridSearcgCV for tuning the hyperparameters and used three-fold cross validation.

The configuration is as follows:

'bootstrap': [True, False],
'max_depth': [10, 20, 30, 40, 50, 60,70, 80,90,100],
'max_features': [2, 3, 4, 5,6,7,8, 9, 10],
'min_samples_leaf': [2,3, 4],
'min_samples_split': [1,2,3,4],
'n_estimators': [300, 400, 500,600,700,800,900]

Hyper-Parameter Tuning: Random Forest

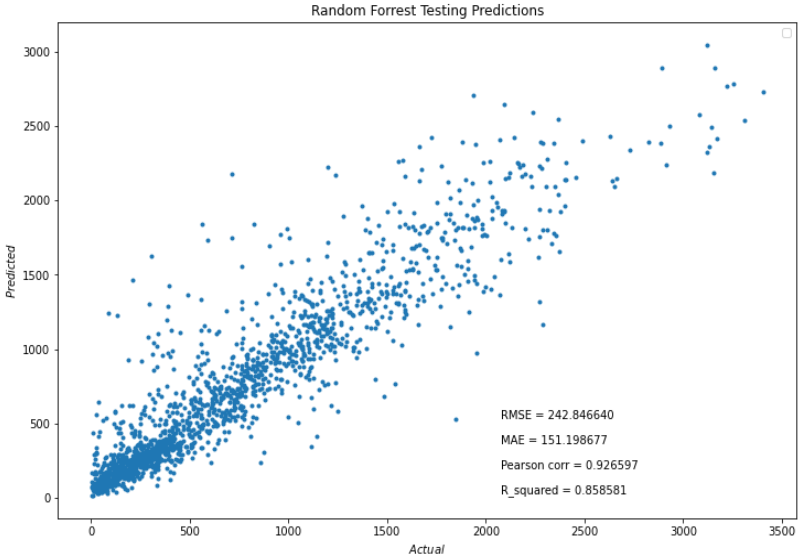
- The best trial configuration after searching all the grid nodes-

'bootstrap': False,
'max_depth': 90,
'max_features': 8,
'min_samples_leaf': 2,
'min_samples_split': 2,
'n_estimators': 500

We trained the RF model again with the optimized parameters.

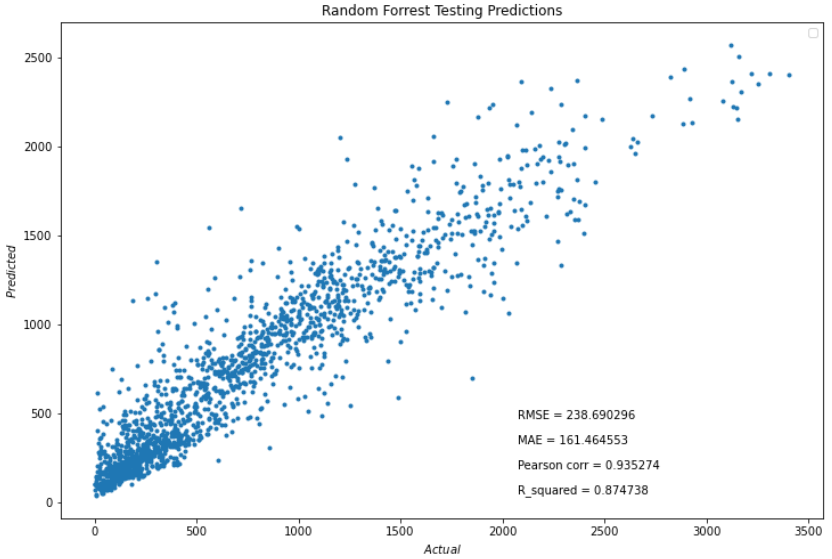
Random Forest

Before Tuning



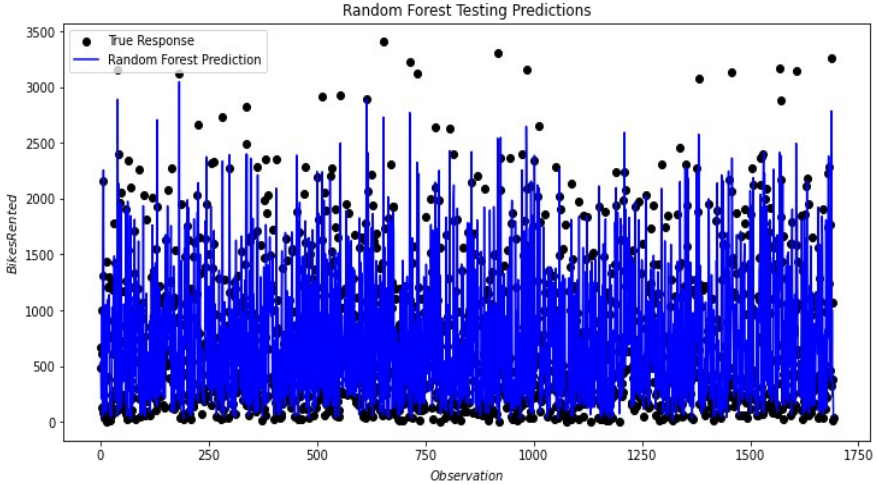
R-Squared = 0.85
RMSE = 151.19
MAE = 242.84

After Tuning

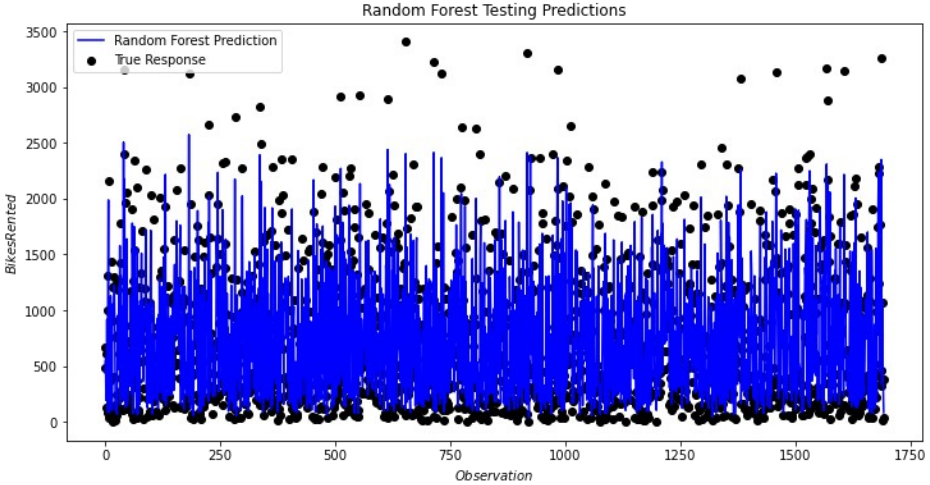


R-Squared = 0.88
RMSE = 161.46
MAE = 238.69

Random Forest

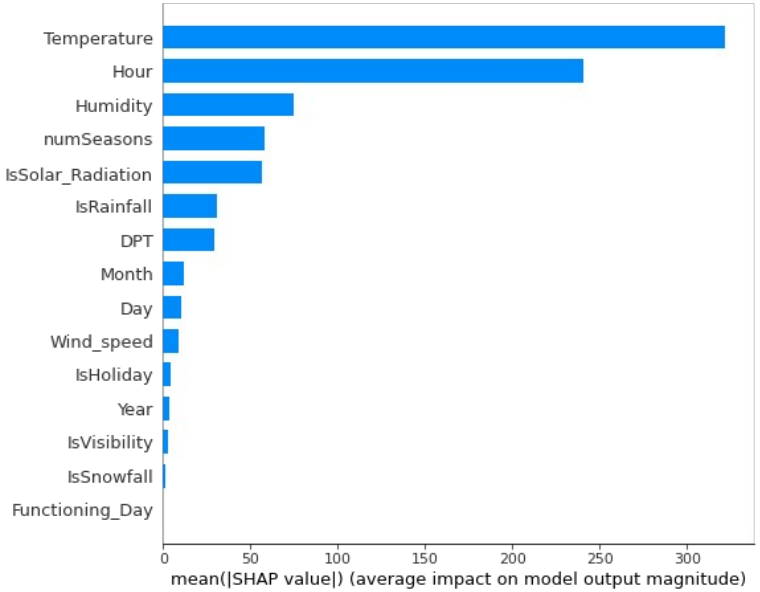


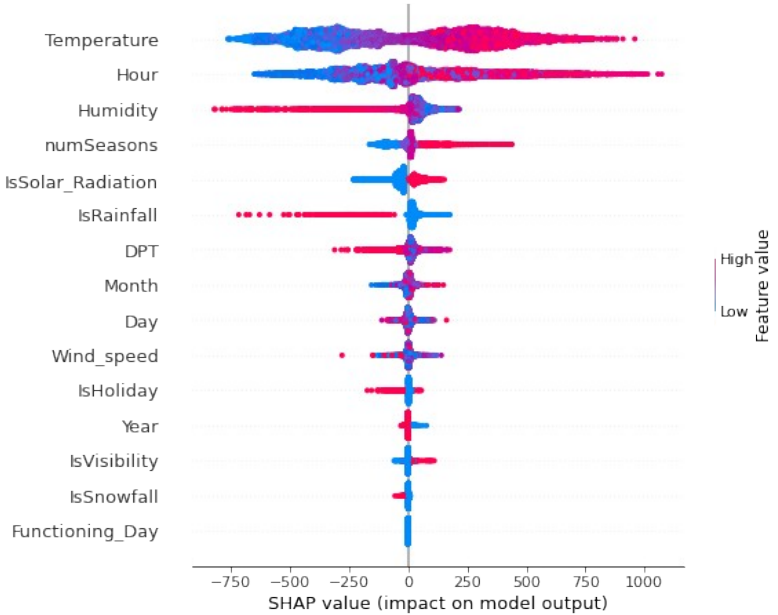
Before Tuning



After Tuning

Random Forest (Feature Importance)





Model Comparison (Table Testing Scores)

Algorithm	*R_Squared*	*RMSE*	*MAE*
Linear Regression	0.7243	338.5612	258.5343
ANN	0.9131	243.7142	185.2045
ANN (tuned)	**0.9612**	**142.1756**	**86.7031**
Random Forest	0.8585	242.8413	151.1945
Random Forest (tuned)	.8821	238.6942	161.4622
XGBoost	0.8833	220.1915	139.3243
XGBoost (tuned)	0.8931	212.105175	136.2714
LSTM	**.9540**	**151.2414**	**87.9315**

Conclusion

- While this data set has high level of variation in the target variable response, we believe that we were able to train an acceptable model.
  - The most accurate model in this case was the tuned ANN model with an R-Squared value of .9612. Last Presentation, we were able to reach an R-Squared value of .9131 with the same network, but different parameters.
- The LSTM method of ML in this presentation also yielded successful results with an R-Squared value of .9540

Requirements

- Is this problem suitable for dealing with by machine learning?
  - Our group believe that this requirement is completed. This dataset is a great set for time-series regression.
- Is the selected machine learning method suitable for the selected problem?
  - Our group believes that this requirement has also been met. From the results (R-squared values of .95 and .96) you can see that while some ML methods perform better then others, the LSTM and ANN models appear suitable for the selected problem.
- Is the machine learning workflow complete?
  - Yes
- Has the objective(s) outlined at the beginning of this project been achieved?
  - Yes
- Does the insights and conclusions follow logically from the results?
  - The results do indeed support the conclusions.

References

[1] UCI Machine Learning Repository. (2020). Seoul Bike Sharing Demand Data Set. [Data file]. Retrieved from https://archive.ics.uci.edu/ml/datasets/Seoul+Bike+Sharing+Demand

[2] Sathishkumar V E, Jangwoo Park, and Yongyun Cho. 'Using data mining techniques for bike sharing demand prediction in metropolitan city.' Computer Communications, Vol.153, pp.353-366, March, 2020

[3] Sathishkumar V E and Yongyun Cho. 'A rule-based model for Seoul Bike sharing demand prediction using weather data' European Journal of Remote Sensing, pp. 1-18, Feb, 2020

Questions?

Thank you!